



7 – Primes + Floor/Ceiling

Week 8 – 7 Oct 2025

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Cheat sheet

Definition

Given any real number x , the **floor of** x , denoted $\lfloor x \rfloor$, is defined as follow

$\lfloor x \rfloor =$ that unique integer n such that $n \leq x < n + 1$.

Symbolically, if x is a real number and n is an integer, then

$$\lfloor x \rfloor = n \iff n \leq x < n + 1.$$

Definition

Given any real number x , the **ceiling of** x , denoted $\lceil x \rceil$, is defined as follow

$\lceil x \rceil =$ that unique integer n such that $n - 1 < x \leq n$.

Symbolically, if x is a real number and n is an integer, then

$$\lceil x \rceil = n \iff n - 1 < x \leq n.$$

Ex. 4.5.21. Prove the statement if it is true and find a counterexample if it is false.
For all odd integers n , $\lceil n/2 \rceil = (n + 1)/2$.

Ex. 4.5.27. Prove the following statement.
For all real numbers x , if $x - \lfloor x \rfloor \geq 1/2$ then $\lfloor 2x \rfloor = 2\lfloor x \rfloor + 1$.

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Bonus Qn: AY24/25 Finals

QUESTION 4.

(c) Prove or disprove the following statements:

(i) For every real number x , $\lfloor -x \rfloor = -\lceil x \rceil$.

Ex. 4.6.14. Prove the statement by contradiction.

For all prime numbers a, b , and c , $a^2 + b^2 \neq c^2$.

SOLUTION . Suppose not. That is, suppose there exist prime numbers a, b , and c such that $a^2 + b^2 = c^2$.

Subtracting b^2 from both sides of the equation gives that

$$a^2 = c^2 - b^2 = (c - b)(c + b).$$

Either $c - b = 1$ or $c - b > 1$ [since $c + b > 0$ and $a^2 > 0$, we must have $c - b > 0$].

Case 1: ($c - b = 1$.)

In this case, because both c and b are prime numbers and the only even prime number is 2, the only possible values for b and c are $c = 3$ and $b = 2$. Then

$$(c - b)(c + b) = 1 \cdot 5 = 5 = a^2,$$

and so $a = \sqrt{5}$. But this contradicts the assumption that a is a prime number.

Case 2: ($c - b > 1$.)

In this case, $a^2 = (c - b)(c + b)$ where both $(c - b) > 1$ and $(c + b) > 1$. Because a is prime, the only positive factors of a are 1 and a , and so, by the unique factorization theorem, the only positive factors of a^2 are 1, a , and a^2 .

Because both $(c - b)$ and $(c + b)$ are greater than 1, the only possibility is that both are equal to a . But this implies that $c - b = c + b$, which implies that $-b = b$, and hence that $b = 0$. This contradicts the supposition that b is a prime number.

Thus a contradiction is reached in both possible cases, and hence the supposition is false and the given statement is true. □

Ex. 4.6.24. Prove the statement in two ways: (a) by contraposition and (b) by contradiction.
The reciprocal of any irrational number is irrational.

SOLUTION . (a) Proof by contraposition:

For all x , if x is irrational, then $1/x$ is irrational

[We want to show $\forall x \in \mathbb{R} \setminus \{0\}$, if $1/x \in \mathbb{Q}$ then $x \in \mathbb{Q}$.]

Suppose x is a real number and $1/x$ is rational. [We must show that x is itself rational.]

Because $1/x$ is rational, there are integers a and b with $b \neq 0$ such that

$$1/x = a/b. \quad (*)$$

Note that $1/x$ is not zero, or else, x is not a number. Therefore, $a \neq 0$. Thus we may solve equation $(*)$ for x to obtain $x = b/a$ where b and a are integers and $a \neq 0$. Hence, by definition of rational, x is rational [as was to be shown]. $\square$

Bonus Qn: AY18/19 Midterm

Let x be a real number. If x^3 is irrational, then x is also irrational.

Ex. 4.7.18. The quotient-remainder theorem says not only that there exist quotients and remainders but also that the quotient and remainder of a division are unique. Prove the uniqueness. That is, prove that if a and d are integers with $d > 0$ and if q_1, r_1, q_2 , and r_2 are integers such that

$$a = dq_1 + r_1 \quad \text{where } 0 \leq r_1 < d$$

and

$$a = dq_2 + r_2 \quad \text{where } 0 \leq r_2 < d,$$

then

$$q_1 = q_2 \quad \text{and} \quad r_1 = r_2.$$

SOLUTION . Suppose that a and d are integers with $d > 0$ and that q_1, q_2, r_1 , and r_2 are integers such that

$$a = dq_1 + r_1 \quad \text{and} \quad a = dq_2 + r_2, \quad \text{where } 0 \leq r_1 < d \text{ and } 0 \leq r_2 < d.$$

[We must show that $r_1 = r_2$ and $q_1 = q_2$.]

Equating the right sides of both equations, we obtain

$$dq_1 + r_1 = dq_2 + r_2,$$

and so

$$r_2 - r_1 = dq_1 - dq_2 = d(q_1 - q_2). \tag{1}$$

Both r_1 and r_2 lie between 0 and d . Thus the difference of $r_2 - r_1$ lies between $-d$ and d , that is,

$$-d < r_2 - r_1 < d.$$

Substituting equation (1) into the inequalities, we arrive at

$$-d < d(q_1 - q_2) < d.$$

Since $d > 0$, dividing throughout the inequalities by d gives

$$-1 < q_1 - q_2 < 1.$$

Since $q_1 - q_2$ is an integer (being the difference of integers) and since it is strictly greater than -1 and strictly less than 1 , we conclude that $q_1 - q_2 = 0$. In other words, we must have $q_1 = q_2$.

Substituting $q_1 - q_2 = 0$ into equation (1), we get

$$r_2 - r_1 = d(q_1 - q_2) = d \cdot 0 = 0.$$

Therefore $r_1 = r_2$ also.

Hence, $r_1 = r_2$ and $q_1 = q_2$, as was to be shown. □

Ex. 4.7.23. Prove that $\sqrt{2} + \sqrt{3}$ is irrational.

Hint: We will use the fact that $\sqrt{2}$ is irrational in our argument.

Ex. 4.7.24. Prove that $\log_5(2)$ is irrational.

SOLUTION . Suppose not. Then $\log_5(2)$ is rational. [We will show that this supposition leads to a contradiction.] By definition of rational,

$$\log_5(2) = \frac{a}{b}$$

for some integers a and b with $b \neq 0$.

By definition of logarithm, $5^{\frac{a}{b}} = 2$. Since $2 > 1$ we must have $5^{\frac{a}{b}} > 1$. This implies that $\frac{a}{b}$ is positive, or else $5^{\frac{a}{b}} \leq 1$. Therefore, we may assume that a and b are both positive. Raising both sides to the b -th power gives

$$5^a = 2^b.$$

Let $N = 5^a = 2^b$. Since $b \neq 0$, we know that $N > 2^0 = 1$. Thus we may consider the prime factorization of N . Because $N = 5^a$, the prime factors of N are all 5's. On the other hand, because $N = 2^b$, the prime factors of N are all 2's.

This contradicts the **unique factorization theorem** which states that the prime factors of any integer greater than 1 are unique except for the order in which they are written. Hence the supposition is false, and so $\log_5(2)$ is irrational. $\square$

Ex. 4.7.28. An alternative proof of the infinitude of the prime numbers begin as follows:

Proof: Suppose there are only finitely many prime numbers. Then one is the largest. Call it p . Let $M = p! + 1$. We will show that there is a prime number q such that $q > p$.

Complete this proof. ($p! = p(p - 1) \cdots 3 \cdot 2 \cdot 1$.)

$$M = p! + 1.$$

Notice that $p!$ has all the different primes inside. We can make the same argument here.

Theorem 4.7.4 Infinitude of the Primes

The set of prime numbers is infinite.

Proof: Suppose not. Then the set of prime numbers is finite, say there are only n many prime numbers.

Then we may list all the prime numbers in ascending order,

$$p_1 = 2, p_2 = 3, p_3 = 5, p_4 = 7, \dots, p_n.$$

Consider the integer $N := p_1 p_2 p_3 \cdots p_n + 1$.

Then $N > 1$, and so by Theorem 4.3.4 (Any integer > 1 is divisible by a prime), N is divisible by some prime number p , i.e. $p|N$.

Also, since p is prime, it must be equal to one of $p_1, p_2, \dots, p_n$.

Clearly $p|p_1 p_2 p_3 \cdots p_n$, hence, by P. 4.7.3, $p \nmid (p_1 p_2 p_3 \cdots p_n + 1) = N$.

Hence $p|N$ and $p \nmid N$, which is a contradiction.